

ARTICLE



Exploring AI-assistance in L2 Chinese writing with standardized assessment tasks

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Abstract

While many studies have addressed the benefits of technology-assisted L2 writing, limited research has delved into how generative artificial intelligence (GAI) supports students in completing their writing tasks in Mandarin Chinese. In this mixed-methods study, 26 university-level Mandarin Chinese foreign language students completed four writing tasks on two separate days. The writing prompts were from the AP Chinese Language and Culture Exam free response questions. Students were permitted to use ChatGPT for assistance with only one of the given prompts each day. After data analysis, writing scores were compared with and without the use of ChatGPT to discern which aspects of ChatGPT students relied on the most. To maintain the validity and reliability of the scores, students' writing samples were independently evaluated by two professional raters, yielding a high degree of inter-rater reliability. The results indicate that utilizing ChatGPT for assistance consistently resulted in higher scores. Additionally, students primarily sought assistance from ChatGPT for vocabulary-related and translation-related questions. Quantitative and qualitative data were used to analyze students' perceptions of using GAI to support the students in completing the writing tasks. The current study seeks to bridge existing knowledge gaps and offer empirical data on how GAI tools, like ChatGPT, can develop students' L2 writing within the Mandarin Chinese language learning context.

Keywords: Generative artificial intelligence; Language assessment; Less commonly taught languages; Mandarin Chinese; Second language writing

Language(s) Learned in This Study: Mandarin Chinese

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Introduction

The landscape of second language (L2) writing instruction is rapidly evolving with advancements in educational technology. Digital tools are increasingly integrated into L2 writing pedagogy, reshaping instructional strategies and approaches (Li et al., 2017; Liaw & English, 2017). These developments, supported by enhanced accessibility and technological advancements, have significantly influenced L2 writing education (Mannion et al., 2019). A pivotal element in this transformation is the advent of AI-powered platforms, exemplified by tools such as Grammarly and Google Translate (Dizon & Gayed, 2021; Tsai, 2019; Yan, 2023). In November 2022, OpenAI introduced *ChatGPT*, which is a GAI model that utilizes leveraging large language models (LLM). ChatGPT, which stands for Chat Generative Pre-trainer Transformer, represents a new milestone in this technological evolution.

ChatGPT, exemplifying the potential use of LLM in education, offers features that not only surpass traditional methods but also introduce novel approaches to teaching and learning (Godwin-Jones, 2022). Investigating the impact and effectiveness of ChatGPT in educational environments, especially in terms

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of innovative teaching approaches, presents a promising field for research. This research examines the role of ChatGPT in aiding Chinese through L2 writing, specifically within the context of learning Mandarin Chinese. We aim to explore the ways in which ChatGPT affects students' writing abilities in high-stakes exams.

By investigating ChatGPT's role in Chinese L2 writing, this research study contributes empirical data that illustrates new insights into the user experience and highlights the broader affordances of integrating AI tools in L2 writing pedagogy. Our study addresses the gaps in the research literature, particularly in understanding the impact of AI on language assessment and the support AI provides to students in the Mandarin Chinese L2 writing context. The study's significance in Chinese language pedagogy illuminates an exploration of how AI can be used in L2 assessment tasks, which addresses the gaps in the literature, as there are no current studies designed with a similar approach to this study. Ultimately, our study aims to delineate ChatGPT's potential benefits and challenges of utilizing ChatGPT in AI-assisted L2 writing tasks. To initially gauge what research has been done with AI-assisted tools, we shift our attention to recent trends in GAI LLM, and how ChatGPT can enhance preparation for language assessment.

Literature Review

AI-Assisted Writing with ChatGPT: Recent Trends

Since the launch and growing popularity of Open AI's ChatGPT in November 2022, a wide range of opinions have emerged regarding the affordances, challenges, and uses of GAI in supporting language learning in classrooms. In particular, research in integrated GAI in high-stakes examinations and classroom-based assessments, is yet to be examined in the research literature. Pedagogically, instructors have been exploring the use of ChatGPT in their instruction, even in content-based courses (Vee et al., 2023). The Writing Across Curriculums (WAC) Clearinghouse recently had an active call for proposals on teaching with text generation technologies, or TextGenEd. The call has transformed the WAC Clearinghouse's special issue on TextGenEd to encourage instructors to share the different types of activities they have used in their current instructional practices. In the special issue, various instructors across different disciplines have shared what they have integrated into their writing pedagogies, which is GAI. In addition, recent publications also include the use and non-use of ChatGPT in experimental settings, with many themes of how GAI technologies can improve writing skills and recognize the importance of AI assistance (Song & Song, 2023; Nghi & Phuc, 2023). With the turn of using GAI technologies, further research can explore how GAI can be used to support second language (L2) writing. GAI technologies can offer valuable assistance for writing assessments, with students possibly utilizing GAI as a resource, such as a language tutor (Godwin-Jones, 2022).

Yan (2023) conducted a study on the impact of GAI assistance in an exploratory setting among undergraduate EFL learners in China. Over one week in a practicum, students used ChatGPT for generative composition, transferred the output to Microsoft Word, and then employed electronic dictionaries for paraphrasing and editing. The findings revealed that GAI tools like ChatGPT significantly aided L2 writing. The findings revealed that while the GAI tool significantly aided L2 writing, it also raised concerns among some learners about plagiarism. The study underscores the need for responsible and ethical use of tools like ChatGPT in language learning. However, research on the use of ChatGPT in high-stakes L2 assessments is emerging.

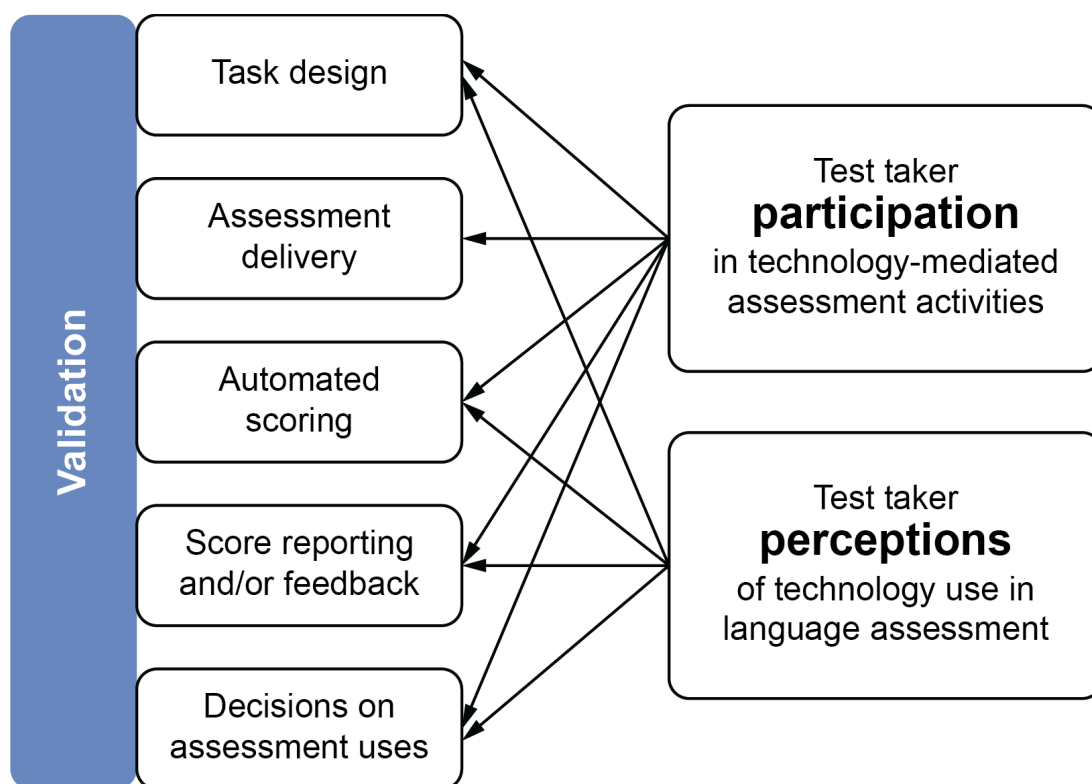
In a study that used ChatGPT in instructing secondary school learners of Mandarin Chinese in an after-school program, X. Li et al. (2023) highlighted the potential of ChatGPT being used as an aide in a specific task design. ChatGPT was introduced and students were taught how to use ChatGPT through International Baccalaureate (IB) writing prompts, followed by five series of lessons that lasted about 45 minutes long, which were followed up by an in-class assessment. If the student passed or failed the assessment, they were asked to use ChatGPT at home and were provided three additional tutorials on how to use ChatGPT as a resource. In the context of the study with secondary school students from

predominantly low socioeconomic status backgrounds, the use of ChatGPT as a GAI technology allowed students to improve their scores on their next attempt at the IB writing prompt which would be used in the in-class assessment. After learning how to prompt ChatGPT, learners expressed how helpful the resource was. The findings of X. Li et al. (2023) inform the current study's rationale to prepare L2 learners of Mandarin Chinese for the Advanced Placement (AP) Chinese Language and Culture exam free-response questions.

Regarding test-taker preparation for standardized language assessments, student engagement should be examined (Jin & Fan, 2023). This aspect pertains to the use of AI technology-mediated language assessment. With the use of generative technologies like ChatGPT to prepare for high-stakes exams like IB and AP tests, it is important to note how widespread GAI tools have not been integrated. Although, automated writing evaluations (AWEs) may currently be provided for IB and AP test takers, depending on the test they take. Jin and Fan (2023) call on the validation of using AI technology-mediated language assessment, which they proposed through the following model.

Figure 1

A Model of Test-Taker Engagement in AI Technology-Mediated Language Assessment



Note. Adapted from Jin & Fan (2023)

In [Figure 1](#), the model illustrates two types of test-taker engagement throughout the phases of developing and validating AI technology-mediated language assessments (Jin & Fan, 2023). The left part of the model suggests that test-takers should be involved in assessment development and use at the phases of task design, assessment delivery, automated scoring of performance-based tasks, score reporting and/or revision of feedback, and making decisions on assessment uses. Validating is a continuous process and should be incorporated into each of the major phases. The model on the right side exemplifies that test-takers can potentially be engaged in two ways: (a) participation in technology-mediated assessment

activities and (b) providing their perceptions of technology use in language assessment. Tapping into their participatory experiences, test takers can form perceptions about technological innovations and offer their perspectives to inform decision-making about the design of the assessment process. Jin and Fan (2023) argue that test takers should be invited to experience and reflect on new technologies, such as ChatGPT. Lastly, test takers should be involved with their perceptions of trying out new technologies through concerns and criticisms.

Artificial Intelligence and Chatbots in the Classroom: Engagement with Mandarin Chinese

There has been a myriad of studies focusing on the ESL and EFL contexts with AI. Most notably, a recent special issue on AI and language learning published by *Language Learning & Technology* focuses on how AI can be used in ESL and EFL classrooms (Warschauer & Xu, 2024). However, there is a lack of studies that focus on less commonly taught languages, such as Mandarin Chinese. To our knowledge, there are few studies on how GAI can assist students in their foreign language writing development in Mandarin Chinese (J. Li et al., 2023; Jiang et al., 2024), and in particular, no studies on high-stakes assessment tasks such as the College Board AP Free Response Questions. J. Li et al. (2023) focused on exploring how ChatGPT could be used in the Chinese language classroom by creating materials for an intermediate low level. The students engaged with ChatGPT through written assessments as a chatbot conversation partner, and what was found from the learners and instructor was that it could be used as a versatile instructional tool, as ChatGPT provided personalized feedback to each student.

Jiang et al. (2024) explored the use of ChatGPT for instructing critical thinking skills in three different Chinese language classrooms. The application of ChatGPT as a chatbot that provided individualized personal feedback allowed students to curate language as different task types affect critical thinking in the three different leveled Chinese language classrooms. Through customized diverse types of tasks with ChatGPT, the learners were able to use critical thinking skills at varying levels. Through the recent findings from research in instructed L2 acquisition, the emerging research from applying GAI and ChatGPT shows innovative efforts in incorporating AI in language learning. In the era of GAI, a more technology-assisted approach can inform a contemporary perspective on how to instruct L2 writing with respect to the Chinese L2 context.

Artificial Intelligence and Mandarin Chinese with L2 Writing Assessments

To determine how ChatGPT impacts writing scores, an investigation into how Chinese language teachers allow students to practice AP writing prompts with the use of GAI tools will be examined in this study. In typical cases, upper-level proficiency students are often assigned to practice for the AP writing prompts by trialing past sample materials officially published by the language assessment company. We utilized the 2021 and 2022 free response questions released by the College Board (2024), which are publicly available to use. To this end, we examine the use of ChatGPT in a less commonly taught language, Mandarin Chinese, and how using ChatGPT as a chatbot may assist with completing the free response questions in a mock test. Building on the existing literature on L2 writing, particularly concerning the use of GAI tools like ChatGPT, our research is trying to investigate the broader implications and potential benefits of integrating these technologies into high-stakes L2 assessments with the following research questions:

1. How does ChatGPT impact students' writing scores with AI-mediated technology assistance?
2. To what extent does ChatGPT assist students in completing their writing tasks?

Methodology

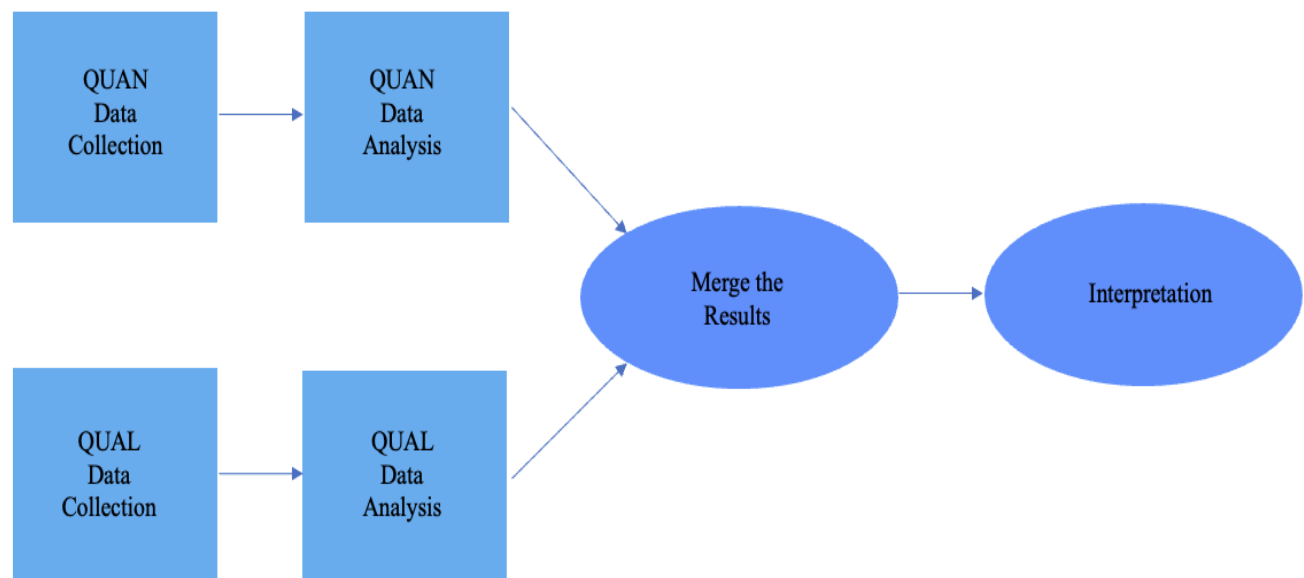
Design Philosophy

The framework adopted for this study is rooted in a convergent mixed-method design, a paradigm that stands out for its foundational and sophisticated nature amidst the vast array of mixed-method designs, as elaborated by Creswell and Crewsell (2022). This design aims to blend findings from qualitative and

quantitative data sources, as Creswell and Crewsell (2022) noted. Creswell and Clark (2018) also highlighted that this design's integration goal is to simultaneously combine data for more comprehensive, expanded insights and interpretations, offering a holistic view, and potentially validating and corroborating the findings. The convergent design we adopted for this study is a one-phase design.

Figure 2

Convergent Design (One-Phase Design)



Note. Adapted from Creswell and Creswell (2022)

Creswell and Creswell (2022) highlighted the use of a convergent design to merge and evaluate both qualitative and quantitative data sources. This methodology involves collecting and independently analyzing these data types before aligning the results to see if they support or contradict each other (see Figure 2). In our study, we utilized this approach, gathering data through a detailed questionnaire with nine multiple-choice and 20 Likert-scale items, plus five open-ended questions for more in-depth insights. We also analyzed ChatGPT logs from students to deepen our qualitative analysis by understanding their specific interactions with the tool.

Participants

To ensure that the AP Chinese writing prompts are appropriate for the participants, this study includes students who are currently learning Mandarin in U.S. colleges, at the third-year level or higher. According to the College Board (2024), the AP Chinese Language and Culture course is equivalent to a fourth-semester college or university course in Mandarin Chinese. Therefore, we assume that students who have completed second-year courses possess the ability to tackle the free response writing prompts from the AP Chinese exam. The participants ($n = 26$) were foreign language learners of Mandarin Chinese. The age range of the participants was between 20 and 25 years old. Out of the 26 students, 16 were in their third year, and 10 were in their fourth year of university. These students represented a variety of majors, including business, humanities, STEM, and diplomacy. Among the participants, 31% had prior experience using GAI, while 69% were using GAI for the first time in this experiment. However, all students indicated their reliance on digital aids to support their language learning, including tools such as Google Translate and digital dictionaries.

Instruments

The experiment spanned two consecutive days and utilized the free response writing prompts from the 2021 and 2022 AP Chinese exams (available on the College Board [website](#)). The AP exam assesses all four language skills, including writing. The writing section of the AP Chinese exam comprises two tasks: story narration and email response (see [Figure 3](#), [Figure 4](#), [Figure 5](#), and [Figure 6](#), published with permission from the College Board). The email response prompts are available in both simplified and traditional characters. The story narration prompt is based on four pictures, assessing whether a student can write a structured, cohesive story. Conversely, the email response prompt aims to determine if a student can appropriately respond to an email written in Chinese. In AP Chinese classes, teachers often have upper-level students practice for the AP writing test using retired sample prompts.

On the first day of our study, students were assigned the 2021 AP writing prompts. They had the freedom to use ChatGPT for the story narration prompt but not for the email response prompt. On the second day, with the 2022 prompts, they could seek assistance from ChatGPT for the email response prompt but not for the story narration prompt. The scope of their inquiries to ChatGPT was restricted to four categories: (a) vocabulary, (b) grammar and structure, (c) sentence translation, and (d) information-related questions (see [Appendix A](#)). Upon completing the tasks on the second day, students filled out a survey to share their experiences working with and without the aid of ChatGPT (see [Appendix B](#) and [Appendix C](#)). The students utilized ChatGPT 3.5 for this study since it was free and available to all users.

We restricted the use of ChatGPT for the following reasons. Firstly, our aim was not for students to generate complete texts through ChatGPT, but to utilize it as a supportive tool in their writing process. Secondly, in interviews with five experienced AP Chinese teachers, it was revealed that students require the most assistance with (a) vocabulary, (b) grammar and structure, (c) understanding sentences at the level of meaning, and (d) gathering the necessary information for crafting responses.

Regarding the development of the survey, we conducted a mock study with ten students and then interviewed them. They shared their experiences and some concerns they had while writing their responses for the story narration and email response tasks. When drafting the post-task survey, we used their feedback as an important source to inform our Likert-scale questions and open-ended questions.

Figure 3

Writing Prompts Day One Story Narration

AP® Chinese Language and Culture 2021 Free-Response Questions

CollegeBoard AP

Writing | Question 1 of 2 00:14:50 Hide Time

Directions: In this task, you will be asked to write in Chinese for a specific purpose and to a specific person. You should write in as complete and culturally appropriate a manner as possible, taking into account the purpose and the person described.

The four pictures present a story. Imagine you are writing the story to a friend. Narrate a complete story as suggested by the pictures. Give your story a beginning, a middle, and an end.

1 2 3 4

Cut Paste Undo Redo Pinyin Simplified Hide Character Count 0

Figure 4*Writing Prompts Day One Email Response (Simplified Characters)*

AP® Chinese Language and Culture 2021 Free-Response Questions

Simplified-character version

CollegeBoard AP®		Switch to Traditional	Help
Writing Question 2 of 2		00:14:22	Hide Time
Directions: In this task, you will be asked to write in Chinese for a specific purpose and to a specific person. You should write in as complete and culturally appropriate a manner as possible, taking into account the purpose and the person described.			
Read this e-mail from a friend and then write a response. 发件人: 加加 邮件主题: 继续学中文 最近好吗? 我高中快毕业了, 毕业以后我还想继续学中文。这两天跟同学聊天的时候说到了上大学后怎么继续学中文。有的说大学里有很多中文课可以上, 有的说别上大学开的中文课, 寒暑假去中国学习中文就行了。这两个选择你觉得哪个更好? 为什么? 另外, 要想学好中文, 你有什么建议?	Cut Paste Undo Redo Pinyin Simplified Hide Character Count 0		

Figure 5*Writing Prompts Day Two Story Narration*

AP® Chinese Language and Culture 2022 Free-Response Questions

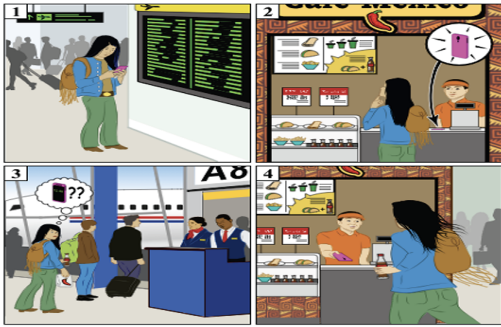
CollegeBoard AP®		Adjust Display	Help
Writing Question 1 of 2		Show Time	
Directions: In this task, you will be asked to write in Chinese for a specific purpose and to a specific person. You should write in as complete and culturally appropriate a manner as possible, taking into account the purpose and the person described.			
The four pictures present a story. Imagine you are writing the story to a friend. Narrate a complete story as suggested by the pictures. Give your story a beginning, a middle, and an end. 	Cut Paste Undo Redo Pinyin Simplified Hide Character Count 0		

Figure 6*Writing Prompts Day Two Email Response (Simplified Characters)*

AP® Chinese Language and Culture 2022 Free-Response Questions

Simplified-character version

CollegeBoard AP®

Switch to Traditional Adjust Display Help

Writing | Question 2 of 2 Show Time

Directions: In this task, you will be asked to write in Chinese for a specific purpose and to a specific person. You should write in as complete and culturally appropriate a manner as possible, taking into account the purpose and the person described.

Read this e-mail from a friend and then write a response.

发件人: 李晓红
邮件主题: 气候环境问题

你好! 我有个问题, 想听听你的看法。今天上课时, 老师讲到了我们这里的气候变化的问题, 比如夏天变得越来越热, 冬天的雪下得越来越少。老师还让我们写一篇作文, 讨论气候的变化对我们生活的影响。你住的地方有哪些气候的变化? 你认为这些变化对我们的生活有什么影响? 谢谢!

Cut Paste Undo Redo Pinyin Simplified Hide Character Count 0

Scoring the Writing Samples

Two AP Chinese raters evaluated students' writing samples, ensuring score validity. Their assessment rigorously followed the [AP Chinese Language and Culture Scoring Guidelines](#) created by the College Board. This approach guaranteed that the evaluation criteria were aligned with standardized educational benchmarks, ensuring consistency and fairness in scoring. The guidelines encompass three key domains: (a) task completion, (b) delivery, and (c) language use. In their assessment process, the raters first conducted a holistic evaluation of each writing sample, followed by a detailed assessment across the specified domains. Consequently, each participant received four distinct scores for each sample, with both the overall score and individual domain scores following a scoring range of 0 to 6. This reflected a comprehensive and methodical evaluation process that adhered closely to the established AP Chinese exam standards.

Data Analysis

To analyze the data, we employed both quantitative and qualitative methodologies. Quantitatively, parametric tests in the R software (version 4.3.1), such as the paired *t*-test, were used to compare scores obtained with and without ChatGPT assistance, as well as to identify any practice effects from the first to the second day of using ChatGPT.

In our coding system, the story narration prompt on day one is labeled as SN1, on day two as SN2, the email response prompt on day one as ER1, and on day two as ER2. Rater one is denoted as R1, while rater two is marked as R2. In this study, students' questions to ChatGPT were categorized into four types: vocabulary (Type 1Q), grammar and structure (Type 2Q), sentence translation (Type 4Q), and information-related questions (Type 4Q). The qualitative data was analyzed thematically (Saldaña, 2021) and will be discussed later under the open-ended responses subsection.

Findings

The Impact of Using ChatGPT versus Not Using ChatGPT on Overall Scores

The data offers insights into how participants performed on story narration and email response tasks, using the assistance of ChatGPT compared to not using it. [Table 1](#) outline the ratings from R1 and R2.

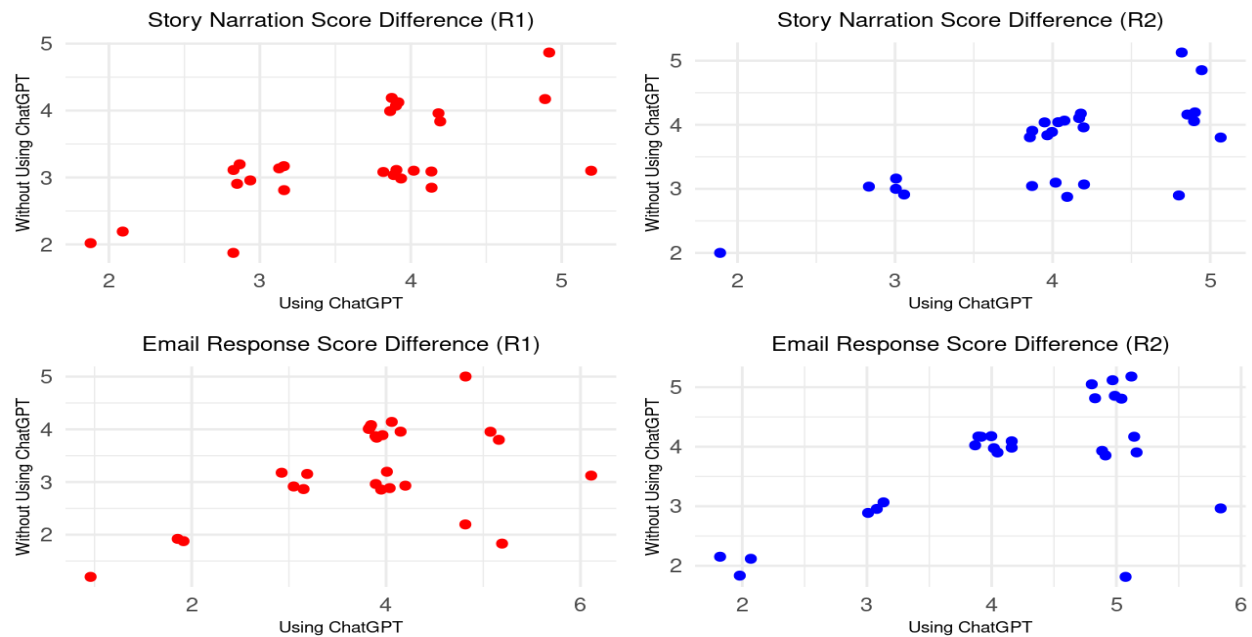
Table 1

Score Narration and Email Response Score Difference Using and Without Using ChatGPT

Rater	Task	ChatGPT	<i>N</i>	Min	Median	Max	Mean	<i>SD</i>
R1	SN1	Yes	26	2.0	4.0	5.0	3.65	0.80
	SN2	No	26	2.0	3.0	5.0	3.23	0.71
R2	SN1	Yes	26	2.0	4.0	5.0	4.04	0.77
	SN2	No	26	2.0	4.0	5.0	3.65	0.69
R1	ER1	No	26	1.0	3.0	5.0	3.23	0.91
	ER2	Yes	26	1.0	4.0	6.0	3.85	1.08
R2	ER1	No	26	2.0	4.0	5.0	3.77	0.99
	ER2	Yes	26	2.0	4.0	6.0	4.15	1.08

Figure 7

Score Narration and Email Response Score Difference Using and Without Using ChatGPT



Story Narration Task (SN)

Based on R1, for the SN task using ChatGPT, participants scored an average of 3.65 ($SD = 0.80$) out of a possible 6, with individual scores ranging from 2 to 5. In scenarios without ChatGPT, the average score decreased slightly to 3.23 ($SD = 0.71$), with scores also between 2 and 5. According to R2, with ChatGPT's assistance, participants achieved an average score of 4.04 ($SD = 0.77$) with a score range from 2 to 5. Without the aid of ChatGPT, they averaged 3.65 ($SD = 0.69$), again with scores ranging between 2 and 5.

Email Response Task (ER)

Following R1, when participants did not utilize ChatGPT for the email response task, they earned an average score of 3.23 ($SD = 0.91$), with scores ranging from 1 to 5. However, when using ChatGPT, their performance improved, averaging 3.85 ($SD = 1.08$) with individual scores ranging from 1 to 6. According to R2, without ChatGPT's support, participants achieved an average score of 3.77 ($SD = 0.99$) with scores from 2 to 5. With the help of ChatGPT, the average score rose to 4.15 ($SD = 1.08$), spanning from 2 to 6. In addition, based on what we observed from Figure 7, there's variability among scores. Using ChatGPT appeared to positively influence the participants' scores, especially noticeable in the email response task. To determine if there is a significant difference between the scores using ChatGPT and without, a paired t -test was conducted, see Table 2 below.

Table 2

Score Differences With and Without ChatGPT Use in Narration and Email Response Tasks

Tasks	t	df	p	MD	CI	d
SN1(R1) vs. SN2(R1)	3.734	25	< 0.01	0.423	0.190 – 0.656	0.556
SN1(R2) vs. SN2(R2)	3.434	25	< 0.01	0.385	0.154 – 0.615	0.533
ER1(R1) vs. ER2(R1)	-3.192	25	< 0.01	-0.615	-1.012 – 0.218	0.621
ER1(R2) vs. ER2(R2)	-2.301	25	0.030	-0.385	-0.729 – 0.040	0.367

Note. * $p = .05$; MD = Mean Difference; CI = 95% Confidence Interval; d = Cohen's d

Based on the result, when comparing the scores for SN1 (using ChatGPT) and SN2 (not using ChatGPT) as rated by R1, the scores were found to be significantly different. The mean score when using ChatGPT was higher by 0.423 units on average compared to when not using ChatGPT ($t(25) = 3.734$, $p = 0.0004$). The 95% confidence interval for this mean difference is from 0.190 to 0.656, indicating that the lower bound of the difference in means is 0.190. Similarly, for the scores rated by Rater 2, there was a significant difference in the scores for SN1 and SN2. The mean score when using ChatGPT was higher by 0.385 units on average ($t(25) = 3.434$, $p = 0.002$). The 95% confidence interval for this mean difference ranges from 0.154 to 0.615.

Contrary to the story narration tasks, for the email response task scores rated by R1, the mean score for ER1 (not using ChatGPT) was lower by an average of 0.615 units compared to ER2 (using ChatGPT) ($t(25) = -3.192$, $p = 0.004$). The 95% confidence interval for this mean difference spans from -1.012 to -0.218, confirming the direction of this difference. Similarly, for the scores rated by Rater 2, ER1 (not using ChatGPT) had a mean score that was lower by an average of 0.385 units compared to ER2 (using ChatGPT) ($t(25) = -2.301$, $p = 0.030$). The 95% confidence interval for this mean difference ranges from -0.729 to -0.040. Therefore, for both story narration and email response tasks, the overall scores are consistently higher with ChatGPT, regardless of the rater.

The Influence of ChatGPT Utilization on Scores Across Three Domains

Based on the above, it was evident that using ChatGPT consistently yielded higher average scores across all three domains of story narration: (a) task completion, (b) delivery, and (c) language use, in comparison

to not using ChatGPT. This trend highlighted ChatGPT's potential to enhance performance in these areas. Similarly, for email response tasks, the data demonstrated that employing ChatGPT consistently led to superior average scores in the categories of (a) task completion, (b) delivery, and (c) language use. This suggested a marked improvement in email response performance when utilizing ChatGPT (see Table 3, Table 4, Table 5, and Table 6). Therefore, irrespective of the task, whether it story narration or email response, the use of ChatGPT consistently resulted in higher average scores across all three domains when compared to its non-use (see Figure 8, Figure 9, Figure 10, and Figure 11).

Table 3

Story Narration Three Domains Scores Using and Without Using ChatGPT (R1)

Domain	Rater	Task	ChatGPT	N	Min	Median	Max	Mean	SD
Task Completion	R1	SN1	Yes	26	2.0	4.0	6.0	4.15	1.08
		SN2	No	26	2.0	3.5	5.0	3.54	0.90
Delivery	R1	SN1	Yes	26	2.0	4.0	5.0	3.73	0.92
		SN2	No	26	2.0	3.0	4.0	3.19	0.63
Language Use	R1	SN1	Yes	26	2.0	4.0	5.0	3.77	0.91
		SN2	No	26	2.0	3.0	4.0	3.27	0.60

Table 4

Story Narration Three Domains Scores Using and Without Using ChatGPT (R2)

Domain	Rater	Task	ChatGPT	N	Min	Median	Max	Mean	SD
Task Completion	R2	SN1	Yes	26	2.0	4.0	5.0	3.92	0.93
		SN2	No	26	2.0	4.0	5.0	3.65	0.80
Delivery	R2	SN1	Yes	26	2.0	4.0	5.0	3.96	0.87
		SN2	No	26	2.0	4.0	5.0	3.58	0.70
Language Use	R2	SN1	Yes	26	2.0	4.0	5.0	3.88	0.95
		SN2	No	26	2.0	3.0	5.0	3.50	0.71

Table 5

Email Response Three Domain Scores Using and Without Using ChatGPT (R1)

Domain	Rater	Task	ChatGPT	N	Min	Median	Max	Mean	SD
Task Completion	R1	ER1	No	26	1.0	4.0	6.0	3.85	1.26
		ER2	Yes	26	0	4.0	6.0	3.92	1.47
Delivery	R1	ER1	No	26	2.0	4.0	5.0	3.50	0.91
		ER2	Yes	26	0	4.0	6.0	3.85	1.41
Language Use	R1	ER1	No	26	2.0	3.5	5.0	3.38	0.80
		ER2	Yes	26	1.0	4.0	6.0	4.00	1.36

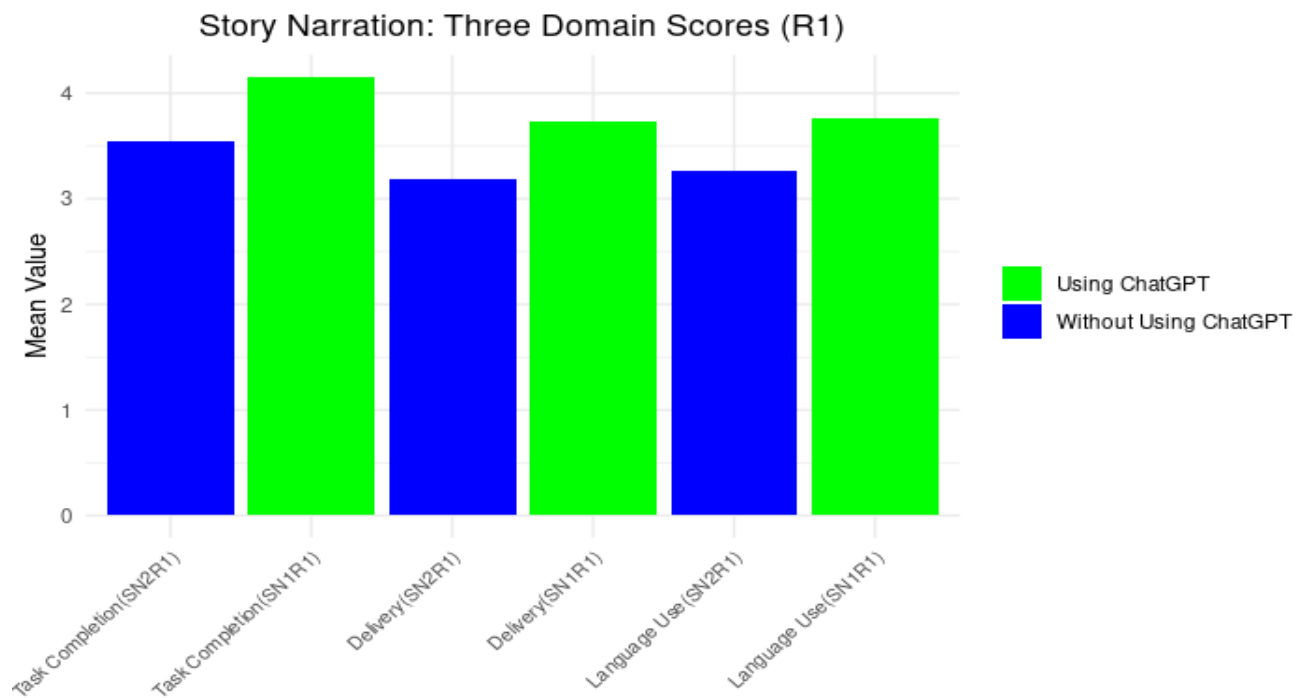
Table 6

Email Response Three Domain Scores Using and Without Using ChatGPT (R2)

Domain	Rater	Task	ChatGPT	N	Min	Median	Max	Mean	SD
Task Completion	R2	ER1	No	26	1.0	4.0	5.0	3.62	1.10
		ER2	Yes	26	1.0	4.0	6.0	3.73	1.31
Delivery	R2	ER1	No	26	2.0	4.0	5.0	3.65	1.02
		ER2	Yes	26	1.0	4.0	6.0	3.73	1.31
Language Use	R2	ER1	No	26	2.0	4.0	5.0	3.54	0.95
		ER2	Yes	26	1.0	4.0	6.0	3.77	1.34

Figure 8

Story Narration Scores in Three Domains Using and Without Using ChatGPT (R1)

**Figure 9**

Story Narration Scores in Three Domains Using and Without Using ChatGPT (R2)

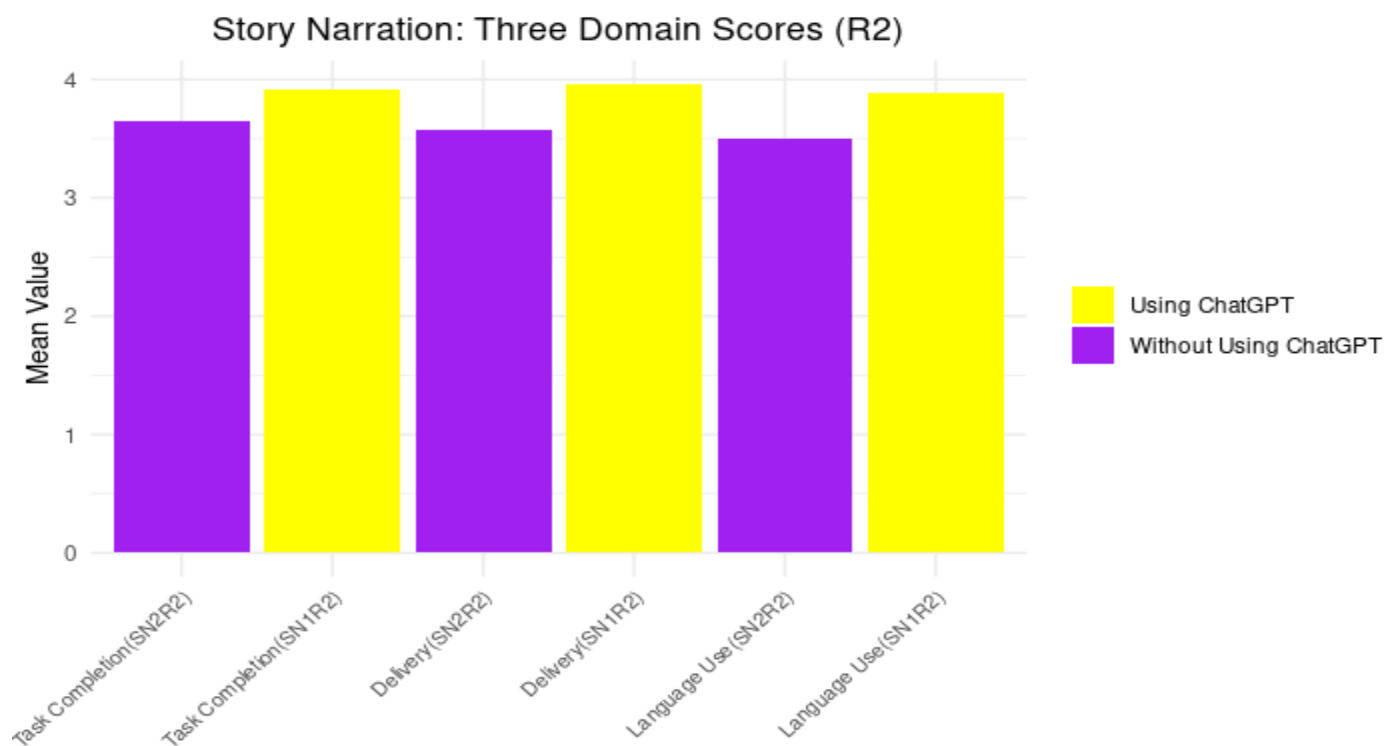


Figure 10

Email Response Scores in Three Domains Using and Without Using ChatGPT (R1)

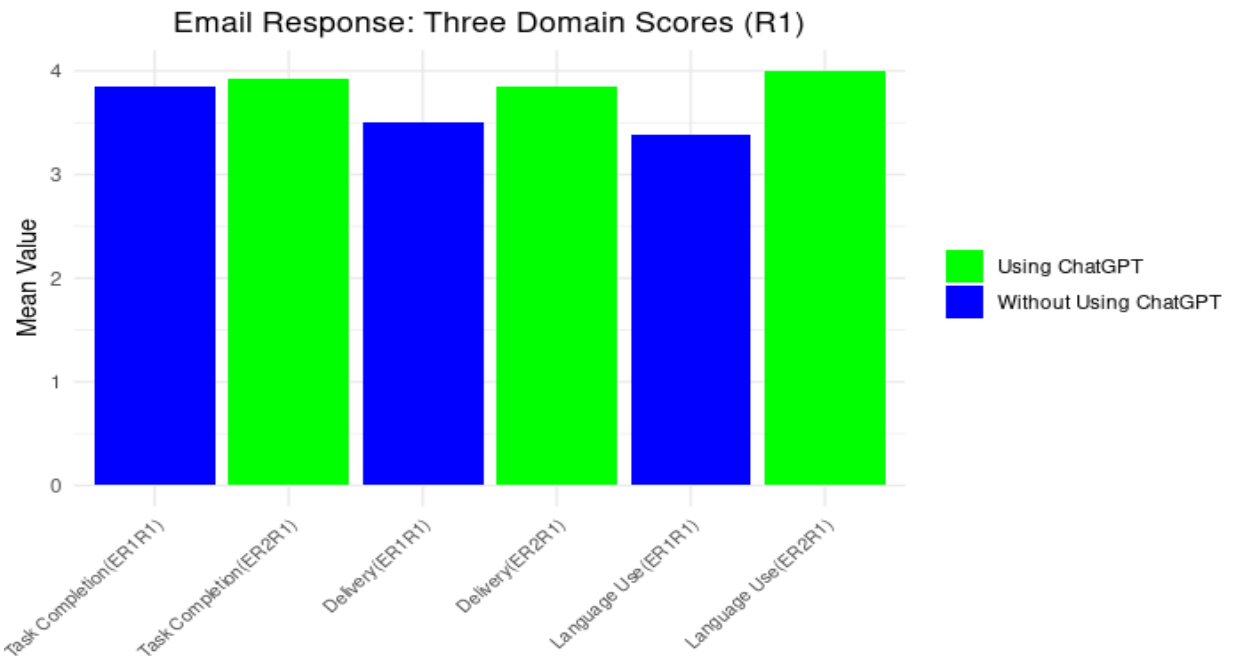
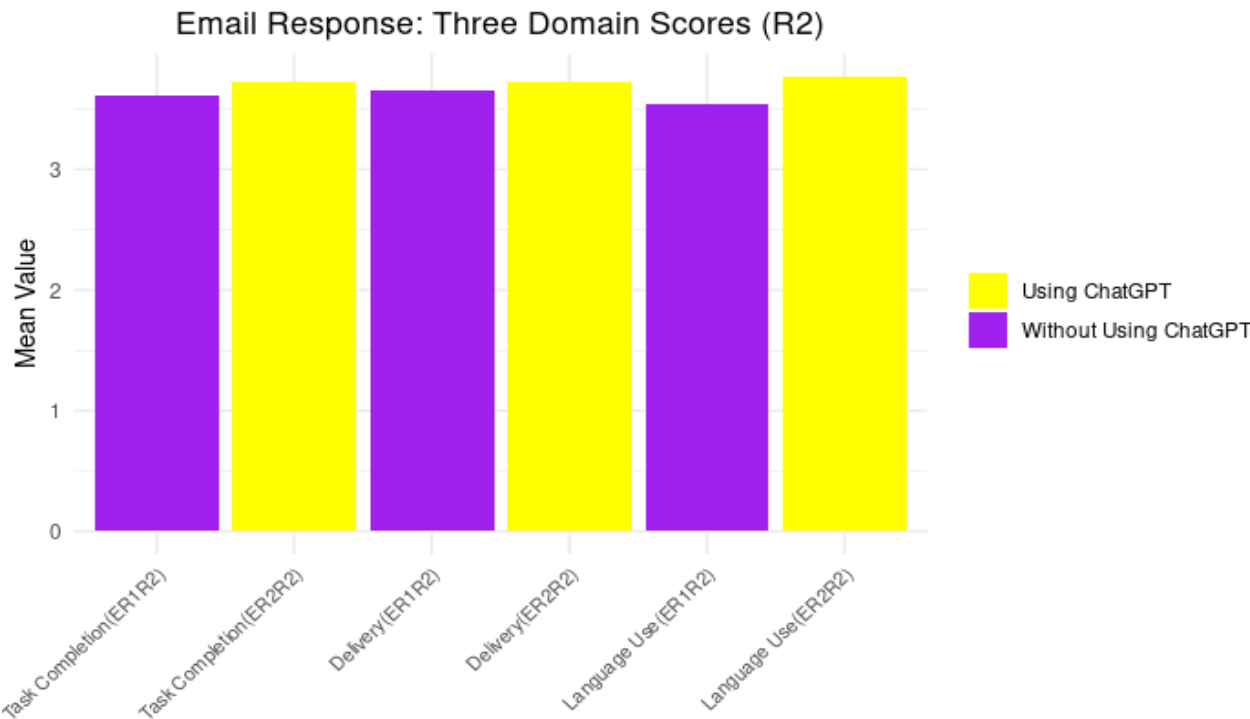


Figure 11

Email Response Scores in Three Domains Using and Without Using ChatGPT (R2)



Score Differences With and Without ChatGPT within the Same Task Type

To determine whether ChatGPT more significantly boosts scores in story narration or email response, the researchers calculated the score difference between using and not using ChatGPT for each task type (see Table 7). This result indicates that using ChatGPT may help to enhance the overall score more significantly for email responses compared to story narration, especially the scores from rater 1. However, when considering the specific domains of task completion, delivery, and language use, using ChatGPT seems to offer more substantial improvements in story narration compared to email responses across both raters.

Table 7

Score Differences: Story Narration versus Email Response With and Without ChatGPT

Rater	Task	Overall	Task Completion	Delivery	Language Use
R1	SN	0.42	0.61	0.54	0.50
R2		0.39	0.27	0.38	0.38
R1	ER	0.62	0.07	0.35	0.50
R2		0.38	0.11	0.08	0.23

Overall Score Impact: With versus Without ChatGPT

To gain a comprehensive understanding of ChatGPT's impact on the writing tasks, the researchers also compared the overall scores obtained with and without using ChatGPT for both task types. Therefore, the researchers added the scores for both SN1 and ER2 (using ChatGPT), and SN2 and ER1 (without using ChatGPT). See Figure 12 for overall scores with and without the use of ChatGPT.

Table 8

Overall Scores Using and Without Using ChatGPT

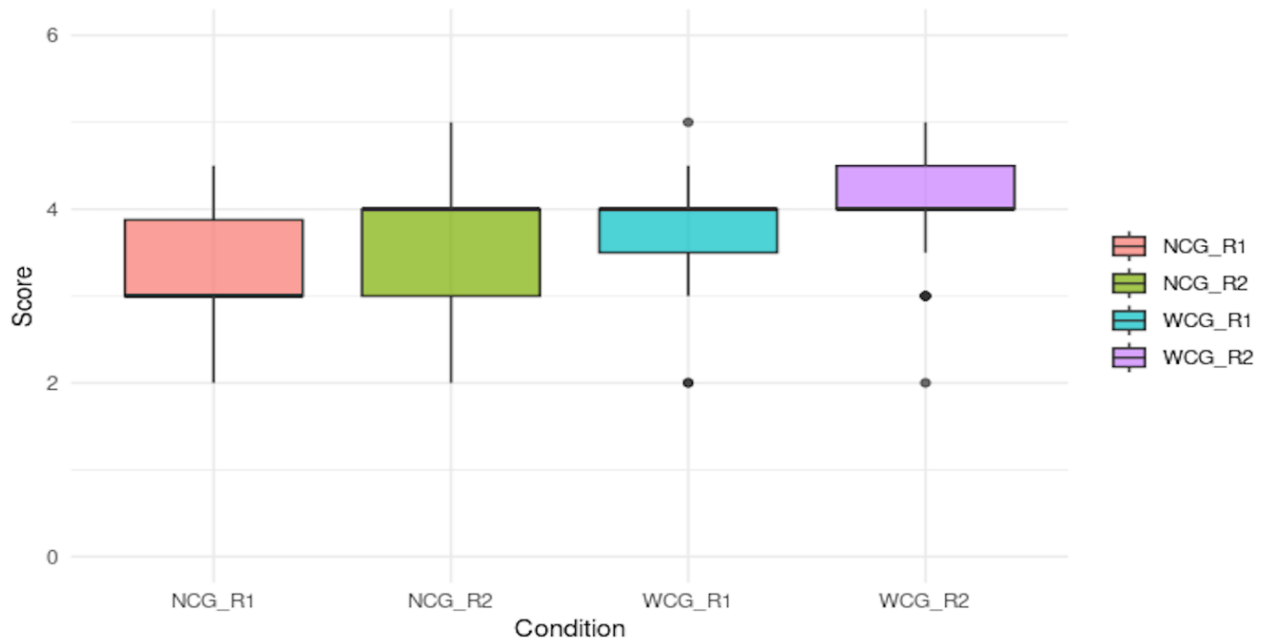
Rater	ChatGPT	Task	N	Min	Median	Max	Mean	SD
R1	Yes	SN1+ER2	26	2.0	4.0	5.0	3.75	0.72
R2			26	2.0	4.0	5.0	4.10	0.79
R1	No	SN2+ER1	26	2.0	3.0	4.5	3.23	0.72
R2			26	2.0	4.0	5.0	3.71	0.79

Table 9

Paired t-test on Overall Scores Using and Without Using ChatGPT

Tasks	<i>t</i>	<i>df</i>	<i>p</i>	MD	CI	<i>d</i>
WCG_R1 vs. NCG_R1	4.758	25	< 0.01	0.519	0.294 – 0.744	0.722
WCG_R2 vs. NCG_R2	4.124	25	< 0.01	0.385	0.193 – 0.577	0.499

Note. WCG_R1 = Overall Scores Using ChatGPT from Rater 1; NCG_R1 = Overall Scores without Using ChatGPT from Rater 1; WCG_R2 = Overall Scores Using ChatGPT from Rater 2; NCG_R2 = Overall Scores without Using ChatGPT from Rater 2; **p* = .05; MD = Mean Difference; CI = 95% Confidence Interval; *d* = Cohen's *d*

Figure 12*Overall Scores Using and Without Using ChatGPT*

Note. NCG_R1 = Overall Scores without Using ChatGPT from Rater 1; NCG_R2 = Overall Scores without Using ChatGPT from Rater 2; WCG_R1 = Overall Scores Using ChatGPT from Rater 1; WCG_R2 = Overall Scores Using ChatGPT from Rater 2

Based on the results from Table 8, Table 9, and Table 10, it is evident that tasks using ChatGPT tend to score higher than those without it. The difference is significant and unlikely to be due to random chance.

The Practice Effect of Using ChatGPT over Two Days

To determine if there was a practice effect on using ChatGPT over two separate days, we conducted a paired *t*-test to compare the scores from the two days of using ChatGPT. Based on the results, there was no statistically significant difference in student performance between SN1(R1) and ER2(R1), or between SN1(R2) and ER2(R2). This suggests that there was no practice effect when using ChatGPT on both day one and day two of this experiment.

Table 10*Practice Effect Between Day One versus Day Two Using ChatGPT*

Tasks	<i>t</i>	<i>df</i>	<i>p</i>	MD	CI	<i>d</i>
SN1(R1) vs. ER2(R1)	-0.795	25	0.434	-0.192	-0.691 – 0.306	0.210
SN1(R2) vs. ER2(R2)	-0.532	25	0.574	-0.115	-0.532 – 0.302	0.117

Note. **p* = .05; MD = Mean Difference; CI = 95% Confidence Interval; *d* = Cohen's *d*

Inter-Rater Reliability

Verifying the inter-rater reliability (IRR) is essential in our study, as it measures the consistency between two raters assessing the same items. In this study, the IRR is defined at both the subject level and task level of the data separately. Based on ten Hove et al. (2022), we propose a method to assess IRR for multilevel continuous ratings provided by two or more raters. ten Hove et al. (2022) explained how IRR coefficients for each level can be derived from these variance components and how they can be estimated as intraclass correlation coefficients (ICC).

Table 11 provides a detailed overview of the ICCs for our study's IRR assessment. It delineates between absolute agreement and relative agreement for various tasks, specifically SN1, SN2, ER1, and ER2. Absolute agreement measures the extent to which raters are in complete concurrence on their ratings across all items. For instance, we observe high consistency in the overall score, with ICCs of 0.94 for SN1 and ER2, but there are lower ICCs for SN2 (0.71) and ER1 (0.63). In terms of relative agreement, which evaluates the consistency in the ranking order by the raters even when their exact scores vary, the ICCs are also high, with SN1 and ER2 showing a score of 0.95, SN2 displaying a 0.81, and SN3 a 0.74. These ICC figures affirm the robustness of the rater agreement within our research, illustrating strong IRR at both the assessed levels.

Additionally, the Pearson correlation coefficient (PCC) is applied to establish the IRR for overall scores and subdomain scores, given its conceptual similarity to ICC's relative agreement. While ICC captures the concordance in ratings, indicating both score consistency and magnitude, PCC evaluates the linear relationship's strength and direction between two sets of continuous data. PCC values can range from -1 for a perfect negative linear relationship, to 1 for a perfect positive linear relationship, with 0 indicating no relationship. The high PCC values observed in our study—0.80 for SN1, 0.82 for SN2 and ER1, and 0.95 for ER2—indicate strong agreement among the raters, which not only complements the ICC results but also provides a quantitative measure of how consistently the two raters have evaluated the students' essays.

Table 11

Intraclass and Pearson Correlation Coefficients

		SN1	SN2	ER1	ER2
Overall Score	AA	0.94	0.71	0.63	0.94
	RA	0.95	0.81	0.74	0.95
	PCC	0.80	0.82	0.82	0.95
Task Completion	AA	0.69	0.58	0.75	0.87
	RA	0.72	0.58	0.79	0.89
	PCC	0.76	0.72	0.83	0.92
Delivery	AA	0.72	0.63	0.79	0.90
	RA	0.73	0.76	0.79	0.91
	PCC	0.73	0.72	0.80	0.96
Language Use	AA	0.78	0.57	0.78	0.88
	RA	0.79	0.62	0.81	0.90
	PCC	0.85	0.76	0.82	0.94

Note. AA=Absolute Agreement; RA=Relative Agreement; PCC=Pearson Correlation Coefficient

Looking at the subdomain scores in Table 11, we find the IRR reflects strong consistency between raters across the different tasks. The task completion domain demonstrates a strong agreement with PCC values of 0.76 for SN1, 0.72 for SN2, 0.83 for ER1, and 0.92 for ER2. Delivery shows PCCs of 0.73 for SN1, 0.72 for SN2, 0.80 for ER1, and an impressive 0.96 for ER2. Language use reveals similarly high

agreement levels, with PCCs of 0.85 for SN1, 0.76 for SN2, 0.82 for ER1, and 0.94 for ER2. These strong PCC values across the board indicate a consistent and robust agreement between raters in all assessed domains.

In conclusion, to address research question 1, the data presented suggests that ChatGPT has a significant positive impact on student writing performance across a variety of tasks. This is evidenced by the notable score improvements in tasks, with an average increase ranging from 0.39 to 0.52 units, observed consistently over two days. The high IRR, demonstrated by the ICC and PCC values, further reinforces the beneficial effects of ChatGPT on enhancing student writing abilities.

Data Analyzed from ChatGPT Logs and Questionnaire

The following sections address research question 2. The study involved analyzing data from participants' ChatGPT usage logs and post-questionnaire responses. This method provided insights into how users engage with ChatGPT and their feedback, offering a comprehensive view of its performance and potential areas for improvement.

Questions Students Asked ChatGPT

As mentioned previously, during their interaction with ChatGPT, students posed questions that fell into four distinct categories: vocabulary (Type 1Q), grammar and structure (Type 2Q), sentence translation (Type 3Q), and information-related queries (Type 4Q). When students posed these questions during a story narration task, they were coded as SNT1Q, SNT2Q, SNT3Q, and SNT4Q, respectively. Similarly, when asked during an email response task, they were labeled ERT1Q, ERT2Q, ERT3Q, and ERT4Q, respectively. Table 12 outlines the distribution of questions asked between each task.

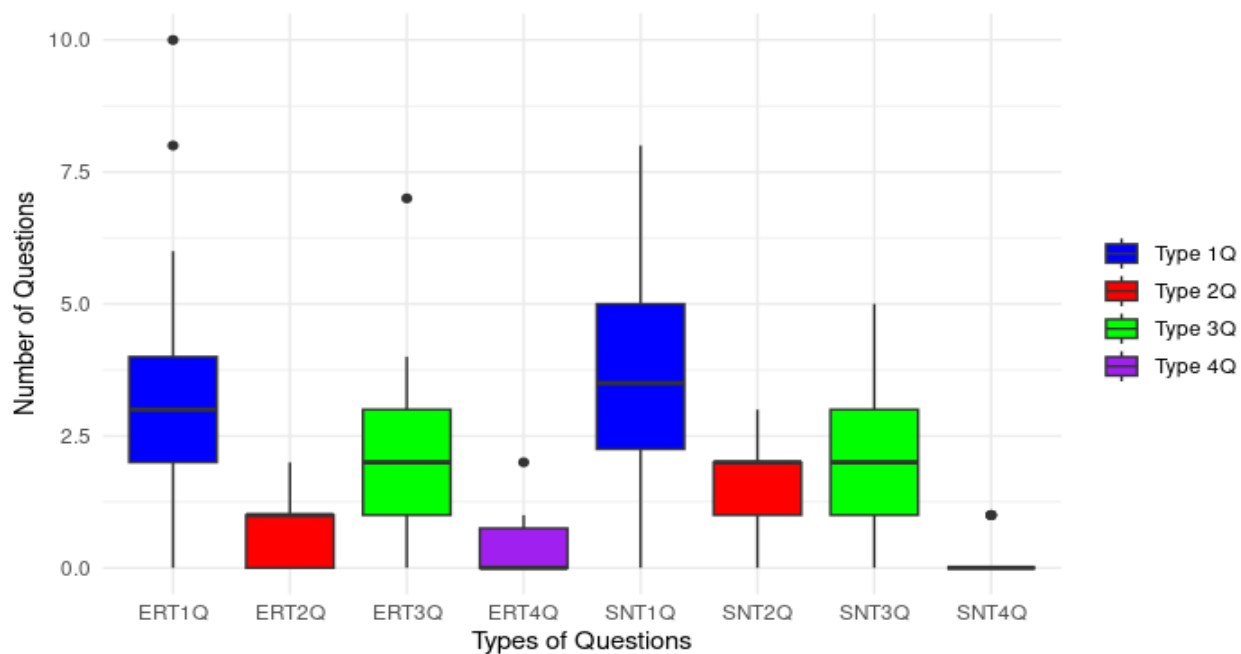
Table 12

Distribution of Questions Asked

Question Type	N	Mean	SD	Median	Min	Max
SNT1Q	26	3.77	2.08	3.5	0	8
ERT1Q	26	3.15	2.39	3.0	0	10
SNT2Q	26	1.58	0.90	2.0	0	3
ERT2Q	26	0.65	0.63	1.0	0	2
SNT3Q	26	2.12	1.56	2.0	0	5
ERT3Q	26	1.88	1.58	2.0	0	7
SNT4Q	26	0.15	0.37	0.0	0	1
ERT4Q	26	0.31	0.55	0.0	0	2

Referencing the provided data from Table 13 and Figure 13, the distribution of questions varied significantly across categories. Vocabulary-related questions, for both story narration and email response tasks, had the highest average counts, with means of 3.77 and 3.15 respectively.

Moreover, students frequently asked ChatGPT about vocabulary, grammar, and sentence translation more than about direct information. Grammar questions showed a wide range in the email task but were less common compared to story narration. Sentence translation queries were moderately frequent, averaging 2.12 in story narration, whereas direct information questions were the least frequent, particularly in story narration, averaging only 0.15. The question types, ranked by their average frequency, were: (a) Vocabulary, (b) Sentence translation, (c) Grammar and structure (overall less frequent but varied in the email task), and (d) Direct information (the least asked). In addition to the quantitative data, here are some examples of the types of questions students asked ChatGPT, taken from the students' ChatGPT logs (see Figure 14).

Figure 13*Distribution of Questions Asked***Table 13***Sample Questions Students Asked ChatGPT*

Questions Type	Sample Questions
Type 1Q	- How do you say “found” in Mandarin? - How do you say “bedroom” in Mandarin - How do you say “relax” in Mandarin?
Type 2Q	- How do you say “to put things under” in Mandarin? - How do you say “instead of doing, but rather” in Mandarin? - How do you say “When...happens, then...” in Mandarin?
Type 3Q	- How do you say “He made me clean up my room” in Mandarin? - How do you say “He was listening to music alone in his room” in Mandarin? - How do you say “I think that global warming is having a significant impact on the recent weather patterns” in Mandarin?
Type 4Q	- How should I open an email in Mandarin? - What are some examples of climate change influencing our lives?

Figure 14

Sample of a ChatGPT Log from Student

D

How do you say "downstairs" in Mandarin?

The Mandarin Chinese word for "downstairs" is "楼下" (lóu xià).





D

How do you say "instead of doing, but rather" in Mandarin?

To say "instead of doing" or "but rather" in Mandarin Chinese, you can use the word "而不是" (ér bù shì). Here's how you can use it in a sentence:

例如 (lì rú): For example,
 我喜欢吃水果而不是巧克力。
 (Wǒ xǐhuān chī shuǐguǒ ér bùshì qiǎokèlì.)
 I like to eat fruit instead of chocolate.





D

How do you say "My dad asks me to clean up my room as soon as possible." in Mandarin?

In Mandarin Chinese, you can say "My dad asks me to clean up my room as soon as possible" as:

我爸爸让我尽快整理我的房间。
 (Wǒ bàba ràng wǒ jǐn kuài zhěnglǐ wǒ de fángjiān.)





Regenerate

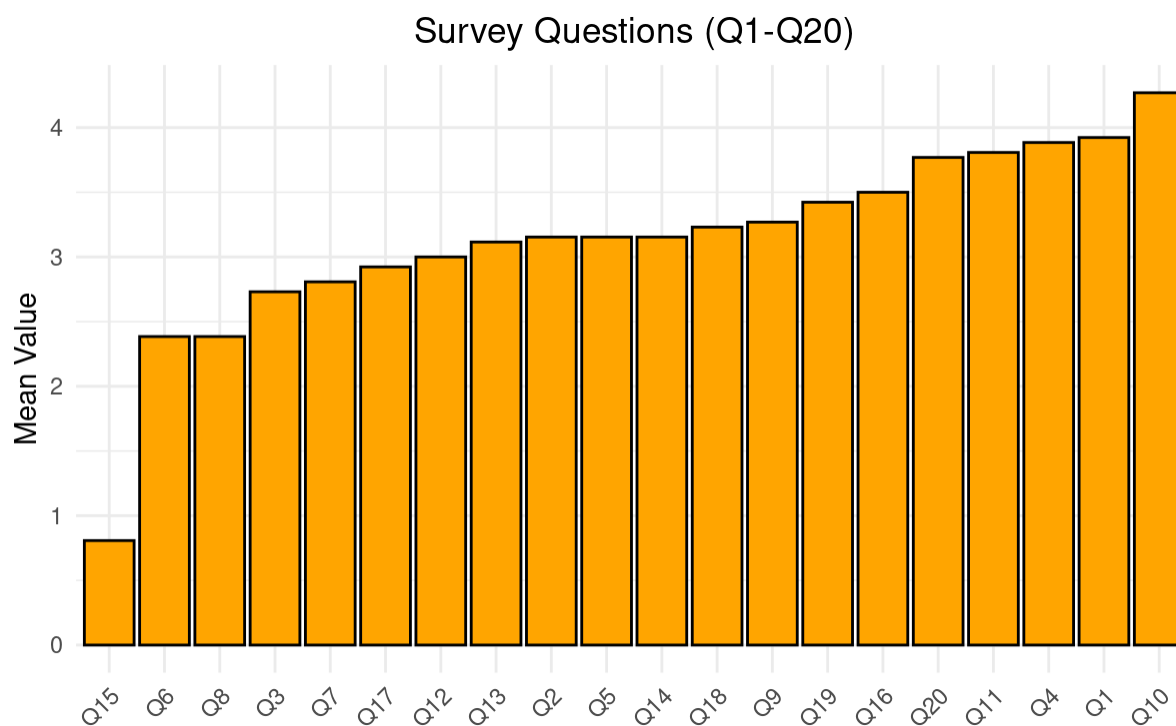
Student Questionnaire Responses

Likert-scale Responses

The survey comprised 20 Likert-scale questions designed to gather students' feedback and experiences using ChatGPT while completing writing tasks. These questions spanned five domains: (a) feedback on vocabulary-related, (b) grammar and structure-related, (c) sentence translation-related, (d) information-related, and (e) general AI usability-related. Responses ranged from 'strongly disagree' to 'strongly agree' and were coded from 0 to 5. To understand students' perceptions of each question, we calculated the mean of the responses to identify the areas where students most frequently agreed or disagreed.

Table 14*Response to Students' Feedback on Using ChatGPT (Likert-scale Questions)*

Question	N	Mean	SD	Median	Min	Max
1	26	3.92	1.23	4.0	1	5
2	26	3.15	1.59	3.5	0	5
3	26	2.73	1.43	3.0	0	5
4	26	3.88	1.18	4.0	1	5
5	26	3.15	1.35	3.0	0	5
6	26	2.38	1.50	2.5	0	5
7	26	2.81	1.65	3.0	0	5
8	26	2.38	1.70	2.5	0	5
9	26	3.27	1.51	4.0	0	5
10	26	4.27	0.67	4.0	3	5
11	26	3.81	1.36	4.0	0	5
12	26	3.00	1.55	3.0	0	5
13	26	3.12	1.51	3.0	0	5
14	26	3.15	1.29	3.0	0	5
15	26	0.81	1.33	0.0	0	4
16	26	3.50	1.39	4.0	0	5
17	26	2.92	1.38	3.0	0	5
18	26	3.23	1.39	4.0	0	5
19	26	3.42	1.50	3.5	0	5
20	26	3.77	1.18	4.0	0	5

Figure 15*Mean Responses to ChatGPT Usage in Mandarin Writing Tasks (Q1-Q20)*

Based on the information from [Table 14](#) and [Figure 15](#), students' perceptions of ChatGPT's utility and impact on their Mandarin writing tasks vary across different aspects.

High Scores (Mean Closer to 5)

Students rated ChatGPT as intuitive and helpful, valuing its ease of use and positive impact on tasks. High-scoring areas include ChatGPT's user-friendliness and usefulness in providing suggestions, as well as a belief in the future role of GAI in language learning.

Moderate Scores (mean around 3)

Scores were mid-range regarding ChatGPT's impact on improving writing skills and enhancing writing speed and naturalness. This suggests room for improvement in areas like context awareness and effectiveness in enhancing writing skills.

Low Scores (mean closer to 0)

Lower scores were noted for technical issues and students' belief that they could complete tasks without ChatGPT or see significant grammar improvement. While technical issues were minimal, enhancing user experience and the tool's impact on areas like grammar could be beneficial.

Overall, while ChatGPT is well-received for its ease of use and role in language learning, improvements are needed in context understanding, skill enhancement, and technical stability. These insights are valuable for refining ChatGPT and similar GAI educational tools.

Open-ended Responses

The open-ended questions were thematically coded and analyzed following Saldaña (2021). Process coding was employed first, focusing on the chronological sequence of students' interactions and inquiries. This enabled us to map out the step-by-step engagement students have with ChatGPT, offering insights into where and how they seek assistance during their writing tasks. Additionally, by employing an in-vivo coding scheme and thematic coding based on responses from the open-ended questions, we highlighted specific words or phrases used by students that captured their experiences or perceptions of ChatGPT's assistance. Descriptive coding is further utilized to sum up the primary topic of segments of data, allowing us to categorize and compare different student interactions and feedback. Through the application of these qualitative coding techniques, we aim to provide comprehensive answers to the research questions, focusing on the practical implications and perceived utility of ChatGPT in the students' writing processes.

Recurring themes were identified in the open-ended questions providing “possible lenses, filters, and angles to consider,” which was why the current study employs an in-vivo coding scheme (Saldaña, 2021, p. 11). We agreed on a set of themes, and employed thematic coding based on the responses from the open-ended questions. There were five themes that we identified from the qualitative data: (a) vocabulary, (b) flow, structure, cohesion, (c) pinyin and Chinese character learning, (d) learning tool, AI tutor, and (e) critical reflection.

The study outlines five themes: the first two focus on students' queries to ChatGPT while working on AP Chinese exam prompts, highlighting its use for immediate language support and as an AI tutor for learning and reflection. The remaining themes explore post-study experiences, showing how ChatGPT facilitated writing tasks, as evidenced in both quantitative and qualitative feedback. This indicates ChatGPT's user-friendliness and its role in aiding students with their writing assignments. These themes are further detailed in the study's analysis.

Vocabulary

The first theme, Vocabulary, related to our research question on how ChatGPT assisted students in completing their writing tasks. This may be connected to their writing scores as seen in Example 1 and 2:

Example 1. I probably would have had issues with specific vocabulary words and structures for things I was thinking of saying, but it probably would have been more indicative of what I could naturally come up with from what I've already studied.

Example 2. ChatGPT was especially good for finding specific vocabulary words at the moment, but there was no retention of that information after I was done writing. It was better for the picture narration for me.

Flow, Structure, Cohesion

Related to the second theme, Flow, Structure, and Cohesion, inform how students used ChatGPT with their writing with the types of questions they asked. Examples 3, 4, and 5 illustrate the second theme:

Example 3. I think it is already quite helpful, because when I asked it to translate sentences from English to Chinese, it translated it and also gave me a breakdown of phrases or characters in the sentence, which cannot be done with Google Translate.

Example 4. ChatGPT was the most useful for looking up unfamiliar vocabulary, however, without giving it full context, I'm not sure how smoothly the suggestions that it gives me fit within the flow of my writing.

Pinyin and Chinese Character Learning

The third theme, Pinyin and Chinese Character Learning emerged from students who were unable to read a character and needed the Pinyin to understand the Chinese characters. Examples 5 and 6 describe this theme:

Example 5. Yes, it kept defaulting to simplified characters despite multiple instructions to only use traditional [characters]. I needed to specify this several times.

Example 6. ChatGPT was very helpful for constructing the email response, as well as identifying meanings and words based on pinyin alone.

Learning Tool and AI Tutor

The fourth theme, Learning Tool and AI Tutor, was informed by the participants indicating that it would be a great resource for language learning in their future Mandarin Chinese courses:

Example 7. As AI improves, I foresee it being used as a "language partner" on a similar level as a native speaker, which allows for remote study of practically any language on one's own time. This would prove especially useful for languages with dwindling numbers of speakers, as it would allow for complete immersion and infinite conversation with another "speaker", given that the AI is mature enough.

Critical Reflection

Finally, the last theme, Critical Reflection, was a theme that we included to share general comments about the affordances of using ChatGPT to assist their Chinese L2 writing:

Example 8. If the presence of AI tools is going to become standard in language learning, then I would hope that they're only used to supplement what students don't know, rather than as a replacement for trying to learn things on their own.

Example 9. The biggest problem I had was to not try and overuse ChatGPT, as I still wanted to preserve some of what I wanted to say in a way that was natural to me and wasn't all elements that the AI had assisted with. The ability to ask how I might say something that I'm thinking in English and the desire to curtail my use of ChatGPT to try and use what I can naturally think of directly in Mandarin usually tripped me up and end up wasting some of the time I had for responses.

Participants in the study had varied experiences with ChatGPT's effectiveness in Mandarin Chinese L2 writing tasks, finding it helpful for grammar and vocabulary searches, including Pinyin for Chinese

characters. Some saw it as a potential language tutor for writing and reading skills. However, there was a consensus on the need for ethical and responsible use of ChatGPT, not relying on it as the sole writing aid, in line with Yan (2023).

In summary, the qualitative data informs the quantitative data, in particular, on the use of how ChatGPT can be used on task preparation for L2 writing assessment in Mandarin Chinese. As ChatGPT continues to gain popularity in language education, the comments from the participants' open-ended questions call for a need in AI literacy education, especially in L2 writing instructional contexts.

Discussion

The data from this study exemplifies the impact of GAI assistance, specifically ChatGPT, on Mandarin Chinese L2 writing proficiency among college students. The quantitative data demonstrates that students' writing scores consistently improved when they utilized ChatGPT for assistance. This enhancement was not limited to just one aspect of writing; it spanned across task completion, delivery, and language proficiency. Such a uniform boost in performance underscores ChatGPT's comprehensive utility in language learning.

A trend to point out from the study is that students predominantly relied on ChatGPT for vocabulary and sentence translation assistance, reflecting a focused use of GAI as a specific tool for particular language elements. This reliance suggests that within the permitted range of vocabulary, grammar and structure, sentence translation, and information-related questions, vocabulary and sentence translation were the areas students found most challenging or in need of support, according to preliminary interviews with the AP Chinese instructors. The less frequent use of grammar, structure, and information queries underlines the selective application of GAI for addressing specific aspects of language learning.

In addition, the findings reflect students' high rating of ChatGPT for its intuitiveness and usefulness in aiding their writing tasks, drawing from their answers in the open-ended questionnaire responses. However, there were areas marked for improvement, such as the tool's contextual understanding from prompts formulated by the participants, as well as the impact of ChatGPT's responses on the participants' writing performance. The range of these perceptions is invaluable for understanding both the strengths and limitations of GAI tools in the educational context, as seen in Yan (2023).

The significance of these findings lies in providing empirical evidence of AI's effectiveness in enhancing L2 writing skills, especially in complex languages like Mandarin. Furthermore, understanding the specific aspects of writing that students turn to GAI for help with predominantly vocabulary and sentence translation, can inform curriculum design and teaching strategies. Educators can integrate GAI tools in a targeted manner, focusing on areas where students are most likely to benefit, such as a pedagogical approach to instructing genre-based writing, as mentioned by Wu (2015).

These insights enable language educators to integrate GAI tools like ChatGPT into their teaching methods, enhancing specific writing tasks and language queries (Baidoo-Anu & Anash 2023; Vee et al., 2023). They are also useful in self-study settings, as described by Godwin-Jones (2009a; 2009b), and guide future GAI tool development for more effective educational technologies (Godwin-Jones, 2012; 2022).

Limitations

The study, focusing on 26 college Mandarin foreign language students, provides insights into GAI's role in Mandarin L2 learning in a controlled academic setting. This initial research suggests how GAI tools like ChatGPT could aid college students' language learning. Future research may involve a more diverse group of participants to broaden our understanding and applicability of these findings to different learner profiles and individual differences. The current focus on specific AP Chinese writing tasks limits the scope. Including a wider array of writing genres in future studies would yield more comprehensive data

on GAI's usefulness in diverse writing situations in Mandarin Chinese L2 learning. Also, counterbalancing the prompts may be preferable, but since this was a classroom-based study, the 2021 and 2022 prompts were used on the first and second days separately to facilitate instruction.

Additionally, future research could explore long-term use to assess how continuous GAI interaction affects language skills, retention, and independent writing development. It should also address the risk of students becoming overly reliant on these tools, potentially hindering their independent learning and critical thinking. Jiang et al. (2024) explore the use of AI and critical thinking and argue that despite language proficiency levels, ChatGPT can assist with written structures used in Chinese for critical thinking prompts.

While technical issues with ChatGPT were minimal, addressing them is crucial for enhancing user experience and reliability. The University of Hawai'i at Mānoa's Language Flagship Technology Innovation Center has developed a [how-to-prompt-ChatGPT website](#) to aid in integrating GAI into language teaching, which could be beneficial for the development of AI literacy.

Given the subjective nature of writing assessment, future studies might benefit from a more diverse panel of raters or automated scoring systems for a balanced evaluation of writing proficiency. Lastly, as GAI technology evolves, ongoing research will be key in adapting educational applications accordingly, as evidenced by resources like the [WAC Clearinghouse's TextGedEd website](#) for example. These adjustments are vital for a comprehensive understanding of GAI tools in language teaching and learning.

Conclusion

In summary, the current study aimed to showcase the teaching and classroom-based assessment of Mandarin Chinese as an LCTL with technology. To this end, we found that the use of GAI tools like ChatGPT relatively supported students in their Mandarin Chinese L2 writing processes on free response writing prompts from the AP Chinese language and culture exam. Applied linguists can fulfill their responsibility to enhance the instruction of LCTLs by harnessing the potential of technology through a dual objective, which involves both the empowerment and preservation of language (Winke & Koné, 2023). As we enter another technological shift in the realm of GAI, we should be open to being adept and creative in serving the needs of students, especially in their writing processes in foreign language education, particularly, for LCTLs.

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Appendix A. The Four Question Types

Vocabulary-related questions

Question Structure: How do you say "the vocabulary you want to ask about" in Mandarin?

Example: How do you say "airport" in Mandarin?

Grammar and structure-related questions

Question Structure: How do you say "the grammar/structure you'd like to know" in Mandarin?

Example: How do you say "from...to" in Mandarin?

Sentence translation questions

You can ask ChatGPT to translate only one sentence at a time. Requests to translate multiple sentences or a paragraph simultaneously will not be accommodated.

Question Structure: How do you say "the sentence you'd like to know" in Mandarin?

Example: How do you say "From my house to the airport takes about 5 hours." in Mandarin?

Information-related questions

For instance, if you're unsure how to answer questions such as "Which places should I visit in Beijing?" or "How should I start an email format?", you can consult ChatGPT for guidance.

Example: What are your recommendations for places to visit in Beijing?

Appendix B. Questionnaire Likert-scale Questions

1. I found using ChatGPT helpful in completing the Mandarin writing tasks.
2. I believe that ChatGPT significantly improved my Mandarin writing skills.
3. I feel more confident in my Mandarin writing skills after using ChatGPT.
4. ChatGPT provided useful suggestions while I was writing in Mandarin.
5. I relied heavily on ChatGPT when writing the email response in Mandarin.
6. I could have written the email response without the help of ChatGPT.
7. I found ChatGPT more helpful during the story narration task than the email response.
8. I believe my Mandarin grammar improved after using ChatGPT.
9. I am likely to use AI tools like ChatGPT for future Mandarin writing tasks.
10. ChatGPT was intuitive and easy to use.
11. I believe that AI will play a crucial role in future language learning.
12. I was able to write more quickly in Mandarin with the help of ChatGPT.
13. The assistance of ChatGPT made my Mandarin writing sound more natural.
14. I believe that my Mandarin vocabulary has expanded after using ChatGPT.
15. I experienced technical issues while using ChatGPT.
16. I felt more comfortable writing the email response with ChatGPT than without.
17. The suggestions provided by ChatGPT always fit the context of my writing.
18. I believe that using ChatGPT improved my score on the Mandarin writing tasks.
19. I would recommend other students to use ChatGPT for Mandarin language learning.
20. Overall, I had a positive experience using ChatGPT in Mandarin writing tasks.

Appendix C. Open-ended Questions

- 1a. Can you share specific instances where you found ChatGPT particularly helpful in your Mandarin writing tasks? For example, was ChatGPT useful for vocabulary, grammar/sentence structure, translations, and/or knowledge level question?
- 1b. Why did you think it was helpful for what you identified in 1a?
2. Did you encounter any problems or issues while using ChatGPT? If so, what were they and how did you overcome them?
3. How do you think your Mandarin writing would have differed if you did not use ChatGPT?
4. In what ways, if any, do you think ChatGPT could be improved to better assist in written Chinese tasks?
5. How do you see AI tools like ChatGPT being integrated into language learning in the future?

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